

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI -590018



MINI-PROJECT (BTC586) REPORT
on

**“AUTOMATED SMOKING ZONE MONITORING AND
ALERTING SYSTEM”**

Submitted in partial fulfillment of the requirements for the award of the degree

BACHELOR OF ENGINEERING

IN

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Submitted by

NAME OF THE STUDENT:

Jonas Benedict V Jose
Kavana N
K Sri Lakshmi
Sandhyaa JK

USN:

1MV24ET402
1MV23ET029
1MV23ET027
1MV23ET051

Under the Guidance of

Ms. Madhu Kumari Ray
Assistant Professor, Dept of ETE Department



SIR M. VISVESVARAYA INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Bengaluru-562157

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SRI KRISHNADEVARAYA EDUCATIONAL TRUST
SIR M VISVESVARAYA INSTITUTE OF TECHNOLOGY

(Affiliated to VTU-Belagavi, Recognized by AICTE and Accredited by NBA & NAAC)
Krishnadevarayanagar, Off Kempegowda International Airport Road, Hunasamaranahalli, Bengaluru – 562 157

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

CERTIFICATE

Certified that the Mini-Project (BTC586) work entitled “AUTOMATED SMOKING ZONE MONITORING AND ALERTING SYSTEM” carried out by Mr. Jonas Benedict V Jose (1MV24ET402), Ms. Kavana N (1MV23ET029), Ms. K. Sri Lakshmi (1MV23ET027), Ms. Sandhyaa J K (1MV23ET051), are the bonafide students of Sir M. Visvesvaraya Institute of Technology, in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Electronics and Telecommunication Engineering of Visvesvaraya Technological University, Belagavi, during the academic year 2025–26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements for the Mini-Project work prescribed for the said Degree.

Ms. Madhu Kumari Ray
Assistant Professor,
Dept. of ETE, Sir MVIT

Dr. Kavitha E.
Professor and Head
Dept. of ETE, Sir MVIT

Dr. M. N.Thippeswamy
Principal
Sir MVIT, Bengaluru

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Jonas Benedict V Jose 1MV24ET402

Kavana N 1MV23ET029

K. Sri Lakshmi 1MV23ET027

Sandhyaa J K 1MV23ET051

DECLARATION

We, students of the 5th semester, hereby declare that the project report titled **“AUTOMATED SMOKING ZONE MONITORING AND ALERTING SYSTEM”** has been presented under the guidance of Ms. Madhu Kumari Ray, Assistant Professor, Dept., of ETE Department of Electronics and Telecommunication Engineering, Sir M. Visvesvaraya Institute of Technology, Bengaluru, in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Electronics & Telecommunication Engineering, Visvesvaraya Technological University, Belagavi, during the academic year 2025–2026. We further declare that this project report has not been submitted previously for the award of any degree or diploma at any institution.

Place: Bengaluru

Date: 08-12-2025

Jonas Benedict V Jose	1MV24ET402
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Kavana N	1MV23ET029
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K. Sri Lakshmi	1MV23ET027
----------------	------------

Sandhyaa J K	1MV23ET051
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ABSTRACT

The Automated Smoking Zone Monitoring and Alerting System is developed to provide an efficient and reliable method for detecting unauthorized smoking activities in restricted areas. Smoking in prohibited zones poses serious health hazards, reduces indoor air quality, and significantly increases the risk of fire incidents. To address these concerns, this project integrates sensor-based detection with real-time monitoring solution. The system employs the MQ-2 gas sensor, which is highly sensitive to smoke and combustible gases, to continuously evaluate air quality within the designated zone. The sensor's analog output is processed by the Arduino UNO, which serves as the central controller of the system. When the smoke concentration surpasses the predefined threshold, the Arduino triggers an immediate audible alert through a buzzer, ensuring quick attention and discouraging smoking behavior. Simultaneously, relevant information such as smoke levels and warning messages is displayed on an I2C 16×2 LCD display, enabling clear and continuous monitoring. The inclusion of an HW battery ensures portability, allowing the system to operate effectively in diverse indoor environments without requiring a fixed power supply. This project presents a low-cost, energy-efficient, and user-friendly solution that can be deployed in public institutions, offices, malls, hospitals, and educational campuses. By automating the detection and alerting process, the system supports the enforcement of smoke-free regulations, enhances safety standards, and contributes to creating cleaner, healthier, and more controlled environments

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